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Going Up

Elevators have been integral to our built environment for over one hundred and sixty years. Since their invention, they have been constantly improved and refined, and are now standard in many buildings around the world. Elevators are machines that are used to transport people and goods between the different floors of a building. This mechanism consists of a cab, which is suspended by a cable. Elevators have been in use since the mid-19th century, and have undergone numerous technological advancements over the years. Primarily, elevators were powered by steam engines and were used in factories and warehouses to move goods. These steam-powered elevators were dangerous, unreliable, and not unsuited for use in buildings. Today, elevators are an integral part of our built environment and are found in almost all modern buildings. They are used not only to transport people between the different floors of a building but also to move goods and equipment. Elevators have made it possible to build taller and taller buildings, allowing us to efficiently and easily move between the different levels of these structures. They have also changed the way we live and work, making it possible for people with mobility impairments to easily access all levels of a building. In this research paper, we will explore the history, technology, and future of elevators in more detail. We will examine how elevators have evolved, from their early steam-powered beginnings to the modern, high-speed elevators of today. Finally, we will look at the future of elevators and the potential for further technological advancements in this field.

The history of elevators can be traced back to ancient Greece, where a primitive version of the device was used to lift water. However, it wasn't until the 19th century that the elevator as we know it today began to take shape. Bellis Mentions elevators “were powered, often steam-operated, and were used for transporting materials in factories, mines, and warehouses” (“The History of Elevators From Top to Bottom” Bellis.1) These elevators were dangerous and unreliable and were not suitable for use in the buildings we have today since their main focus was to transport goods and not humans. In 1853, Elisha Otis designed the first practical elevator, which was equipped with a safety mechanism that prevented the car from falling if the rope broke. This innovation made it possible for buildings to rise much greater in height, leading to the construction of the first skyscrapers. Prior to the invention of the elevator, buildings were typically limited to a few stories due to the difficulty of climbing stairs.

The safety of elevators has been a concurrent question in the minds of those that put elevators in their buildings. According to the Consumer Product Safety Commission, there have been forty-one deaths related to elevators between the years 2018 and 2021. In just a matter of three years, forty-one people have died and more than nineteen thousand were injured in the United States alone. (“CPSC Recalls More Than 100,000 Residential Elevators in 2021-2022” CPSC) These numbers are astronomically high considering how far humans have advanced technologically. A scholarly article presents a smart elevator management system that combines Building Information Modeling (BIM) technology, internet technology, embedded technology, and software management technology with an elevator intelligent management system. This is one way to reduce the deaths and injuries that are happening from elevators. Ziliang uncovers “a smart elevator management system based on ‘BIM + IOT’ ” to make elevators safer. This system is designed to monitor the status and operation of elevators, enable multi-party joint rescue

efforts, and facilitate information sharing among government agencies, enterprises, property managers, and maintenance providers. By doing this every party that is affiliated with elevators is able to implement the new findings for elevator safety. Ziliang Articulates, The system consists of hardware and software components that aim to improve the safety and intelligence of elevators by enabling functional coupling between elevators and intelligent buildings. ("Ziliang, Building Elevator Safety Monitoring System Based on the BIM Technology. Journal of Physics) This research and development help build a working foundation to further improve the safety of elevators.

In Germany, a company called ThyssenKrupp is developing a new type of elevator that moves not just up and down, but also sideways. Engineer Marcos Yetta Goes on to say, “The idea of developing a cable-less elevator has been around for a century, but we are now taking this step and breaking new ground.”(Yetta, Skyscraper elevators: up, down and sideways | DW News) They are finally taking action on their ideas and have started working on what could be the future of elevators. This type of elevator, known as a Multi elevator, is designed to serve multiple floors and can move in different directions within a building. The Multi elevator uses a linear motor to move along tracks, allowing it to travel horizontally as well as vertically. This allows it to serve multiple floors without the need for separate shafts or stops, making it more efficient and space-saving than traditional elevators. The Multi elevator is still in the development phase, but it has the potential to revolutionize the way we move within buildings. It would allow buildings to be designed in new and innovative ways, as the need for separate elevator shafts on each floor would be eliminated. It would also make it easier for people to access different parts of a building, as the Multi elevator could move horizontally as well as vertically. In addition, the use of a linear motor would make the Multi elevator more energy efficient than traditional elevators.

Overall, the development of the Multi elevator is an exciting development in the field of elevator technology and has the potential to greatly improve the way we move through buildings in the future.

Elevators have come a long way since their invention in the 19th century. From their early steam-powered beginnings to the modern, high-speed elevators of today, elevators have undergone numerous technological advancements. They are an integral part of our built environment and have greatly changed the way we move through buildings. With the current technology in today's day and age, there isn't a limit on the potential of what elevators could do.

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